CAMPUS CAFE ORDERING SYSTEM

Java OOP Laboratory Exercise Prepared

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Section: D

Date: 11/10/2026

1.Introduction

The Campus Cafe Ordering System is a Java console program that processes customer orders of up to three menu items. It supports beverages, pastries, and meals. Each menu item calculates its price based on its own options. The system can apply a student discount, calculate a voucher deduction, track customer loyalty points, and print an order receipt.

2. Object-Oriented Programming Concepts

Classes and Objects. The program uses seven classes: MenuItem, Beverage, Pastry, Meal, Customer, Order, and CafeApp. CafeApp creates customer, menu-item, and order objects and uses them to process sample transactions.

Abstraction. MenuItem is an abstract class that contains common menu-item information and declares computePrice() as an abstract method.Beverage, Pastry, and Meal provide their own implementations of this method.

Inheritance. Beverage, Pastry, and Meal extend MenuItem. They reuse common item information and methods from the parent class.

Encapsulation. The classes use private and protected instance variables instead of public fields. Getters and setters provide controlled access, and validation prevents invalid values such as an unsupported beverage size or a negative number of extra sides.

Polymorphism. Order stores menu items using the MenuItem type. When the program calls computePrice() or getDescription(), Java uses the overridden method belonging to the actual item type.

Constructors and Overloading. The classes use constructors to initialize objects. Customer demonstrates constructor overloading. The program also uses overloaded methods for adding loyalty points and computing an order total.

Conditional Statements. if and if-else statements are used to validate inputs, determine item prices, check the three-item order limit, and apply discount and voucher calculations.

3. Program Features

• Supports up to three menu items in each order.

• Supports beverages, pastries, and meals.

• Calculates prices according to each item's selected options.

• Applies a 10% student discount when applicable.

• Supports a voucher deduction without allowing the total to become negative.

• Tracks and displays customer loyalty points.

• Prints receipts for customer orders.

• Demonstrates input validation and rejection of a fourth item.

4. Challenges and Solutions One challenge was organizing the project into separate Java class files and making the classes work together. This was addressed by placing each class in its own file and using MenuItem as the common parent class for the different menu-item types. Another challenge was running the program in IntelliJ IDEA. The project was organized with CafeApp as the main class, and the program was successfully run. Input validation and the three-item order limit were also tested through the sample program.

5. Testing The CafeApp main method creates two customers, several menu items, and two orders. The program tests an invalid beverage size and a negative extra-side value, then attempts to add a fourth item to an order to demonstrate the item limit. It prints both customer receipts and calculates the second order's total with a voucher. The submitted test run completed with "Process finished with exit code 0," indicating that the program ran successfully.

6. Conclusion The Campus Cafe Ordering System demonstrates how Java object-oriented programming concepts can be used to build a simple ordering application. The project uses classes, objects, inheritance, abstraction, encapsulation, polymorphism, constructors, method overloading, and conditional statements. It also shows how separate classes can work together to process orders and display receipts.

OUTPUT

"C:\Program Files\Eclipse Adoptium\jdk-25.0.5.8-hotspot\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.3\plugins\java\lib\idea_rt.jar=60709" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -classpath C:\Users\Joven\IdeaProjects\Torres_CafeSysstem\out\production\Torres_CafeSysstem CafeApp

Testing invalid inputs:

Invalid beverage size!

Extra sides cannot be negative!

Iced Coffee added to order.

Chocolate Muffin added to order.

Chicken Rice added to order.

Adding a fourth item:

Order can only contain 3 items!

Milk Tea added to order.

Cheese Bread added to order.

Burger Meal added to order.

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CAMPUS CAFE RECEIPT

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Order ID: ORD-001

Customer: Matt

Iced Coffee Size: M PHP 95.00

Chocolate Muffin PHP 50.00

Chicken Rice PHP 120.00

Subtotal: PHP 265.00

Student Discount: PHP 26.50

Total: PHP 238.50

Loyalty Points: 4

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CAMPUS CAFE RECEIPT

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Order ID: ORD-002

Customer: Malia

Milk Tea Size: L PHP 115.00

Cheese Bread (Heated) PHP 55.00

Burger Meal (Combo) PHP 210.00

Subtotal: PHP 380.00

Student Discount: PHP 0.00

Total: PHP 380.00

Loyalty Points: 7

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Total with PHP 20 voucher: PHP 360.0

Process finished with exit code 0